Optimized GNSS Network Station Selection to Support the Development of Ionospheric Threat Models for GBAS

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ABSTRACT

Extremely large ionospheric spatial gradients present potential integrity threats to the users of global navigation satellite systems (GNSS) augmentation systems. Thus, these ionospheric anomalies need to be monitored by ground reference stations and users must be alarmed with feasible time-to-alerts. The long-term ionospheric anomaly monitoring (LTIAM) tool has been developed to monitor ionospheric behavior continuously over the life cycle of ground-based augmentation systems (GBAS) and to build ionosphere threat models for all regions where GBAS will be fielded in the future. The use of poor-quality GNSS data degrades the accuracy of ionospheric delay estimates and thus the performance of the resulting threat models.

Previous work developed a comprehensive method of GPS data quality determination to identify stations that are poor according to multiple quality metrics. This method eliminates almost 90% of spurious or false gradients when applied to the continuously operating reference stations (CORS) in the conterminous U.S. (CONUS). However, use of this method typically eliminates more stations than necessary because it determines the thresholds for each quality parameter independently without considering the interrelations between the parameters.

This paper develops an optimized method to select well-distributed, high-quality stations for a GNSS network. Thresholds which maximize the elimination of spurious gradients while minimizing unnecessary station removals are established. This method achieved an 89% reduction of faulty anomaly candidates while discarding only 16% of the more than 1500 CORS stations. Because this method is also used to remove redundant stations in dense regions with a desired baseline constraint, the benefit when using it is that it reduces the processing time of LTIAM. These algorithms are also tested using data collected from other GNSS reference station networks to verify the performance of the proposed method.

1.0. INTRODUCTION

The long-term ionospheric anomaly monitoring (LTIAM) tool is an automated software package developed to build ionospheric anomaly threat models from observed events by analyzing past data and to continuously monitor ionospheric behavior over the life cycle of GBAS [1, 2]. The LTIAM tool will be used to verify the current threat model over the life cycle of the system and update it if necessary. As the current solar cycle approaches its maximum in 2013-2014, this is of particular importance over the next few years.

The LTIAM software can observe ionospheric gradients over short-baseline distances of 5 – 40 km using data automatically collected from the CORS network, which had over 1500 stations as of 2011 in the CONUS and a few other countries compared to about 400 stations prior to 2004. As the total number of stations increases, the observability of ionospheric behavior improves but the number of stations with poor GPS data quality also increases. Thus, while automated measurement screening is included in LTIAM, the use of poor-quality data produces many cases of faulty anomaly candidates and thus burdens the process significantly.

Previous work developed a comprehensive means of determining the quality of GPS data to identify stations with poor quality data which produce faulty anomaly candidates. The use of this one-dimensional one-step search method with CORS stations in CONUS eliminates nearly 90% of spurious or false gradients. However, thresholds to remove poor stations were established independently of each quality metric without considering the interrelations between parameters. Because the violation of any one threshold results in an exclusion, this method eliminates more stations than necessary.

This paper develops a new station-selection algorithm to establish thresholds which maximize the elimination of spurious gradients while minimizing unnecessary station removals. In this algorithm, a set of thresholds for the quality parameters is determined using a two-dimensional iterative concept which searches for the most troublesome station and at the same time the most efficient quality parameter. Thresholds are derived for each of these metrics, and stations which exceed the determined threshold are excluded from LTIAM measurement process unless they are recovered by a secondary check of their location. Stations whose locations for observing the ionosphere are relatively more important are retained despite their low data quality.

Regardless of the data quality, a large number of stations which are geographically redundant in dense regions bring an excessive computation load to the automated post-processing of data by LTIAM. As long as an area of interest is covered by a sufficient number of stations with separations of less than a desired baseline distance, other nearby stations in the region which do not improve the geographic observability of the ionospheric behavior are redundant and can be removed even if they provide high-quality data. This paper therefore proposes a method to select a subset of stations which are geographically important among good-quality stations. This type of subset of stations which are evenly distributed and which maintain an area covered by stations with a desired baseline distance will allow a reduction of the processing time of LTIAM and consequently help to build ionosphere threat models. In this paper, Section 2.0 illustrates the effect of poor-quality data; Section 3.0 explains the optimized GNSS network station-selection methodology in detail; Section 4.0 and Section 5.0 show the results of applying this method to CORS stations in CONUS and other GNSS reference networks, respectively, and Section 6.0 concludes the paper.

2.0. POOR CORS DATA QUALITY

This section investigates the effect of GPS data collected from stations with poor data quality on the ionospheric delay and gradient estimation. Figure 1 shows real example of poorly sited CORS stations in CONUS. The data from these stations which are located next to a solar panel, in a bush, and even in between towers may produce erroneous estimates of ionospheric delays, and consequently unusually large ionospheric gradients due to corrupted raw measurements.

Figure 1. Real Eaxmple of Poorly Sited CORS Stations in CONUS

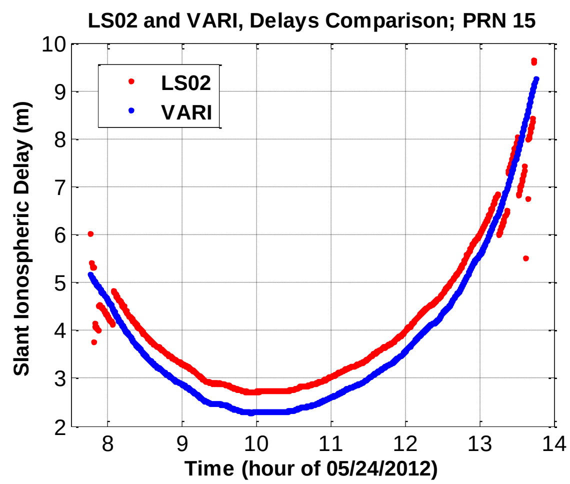

Figure 2. Dual-frequency Slant Ionospheric Delay Estimates for Stations LS02 (Poor Quality Data) and VARI (Good Quality Data)

Figure 22 shows the slant ionospheric delays observed from two nearby stations LS02 and VARI while they tracked PRN 15 on 24 May 2012. LS02 is a good example of a station with poor GPS data quality. From the many fragments of ionospheric delay estimates from LS02 (red), it is evident that its carrier-phase measurements are corrupted by numerous cycle slips, resulting in outliers and short arcs of valid measurement and outliers. LS02 (red) also held a small amount of data compared to the normal station VARI (blue). Figure 33 shows the estimated ionospheric spatial gradients between LS02 and VARI. The ionospheric-delay leveling errors due to the short arcs from LS02 are observable at each end of the curve, and the large gradients due to the excessive cycle slips on LS02 are evident in the center of the curve. This example illustrates how poor data quality degrades the accuracy of ionospheric delay estimation and can produce extremely large ionospheric gradients that are not real.

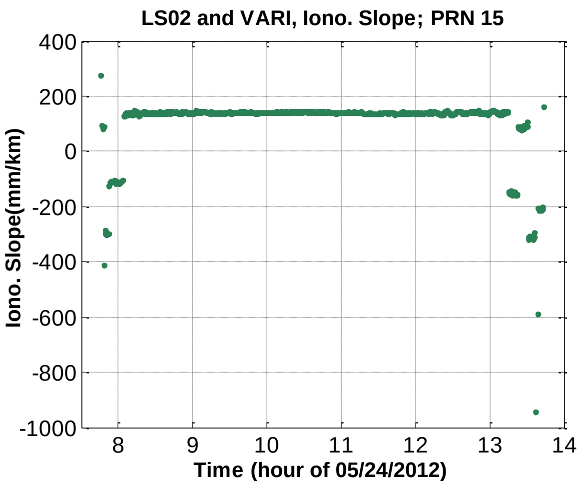

Figure 3. Ionospheric Gradient Estimates Corrupted by Poor Quality Data from Station LS02

3.0. DATA ANALYSIS METHODOLOGY

The methodology for selecting optimized GNSS network stations with well-distributed, high-quality data is composed of three steps: measuring GPS data quality information using several metrics, determining the thresholds of each of these metrics to remove stations with poor-quality data, and re-examining not only tentatively excluded stations but also redundant stations in a dense area considering their geographical locations. First, data quality information is obtained by processing the RINEX file collected at each station through a series of data-quality-measurement algorithms [3]. Second, stations with sufficiently poor-quality data are selected based on whether the quality parameters of the station exceed established thresholds. Third, to observe anomalous ionospheric gradients throughout the area, the remaining stations should cover throughout the area with separations of less than 40 km to the degree possible. To meet this criterion in regions with relatively few stations, some degrees of data quality may need to be sacrificed. Therefore, we examine the geographical contribution of stations selected for removal. Stations whose location increases the geographical observability of ionospheric behavior are restored despite their poor data quality. On the other hand, a good-quality station may be geographically redundant if located in a dense area. To select a subset of stations which are geographically important among good-quality stations, Stations whose location do not help in terms of the observablility of ionosphere and only increase computational load are removed despite their good data quality. The details of each step are described in the following subsections.

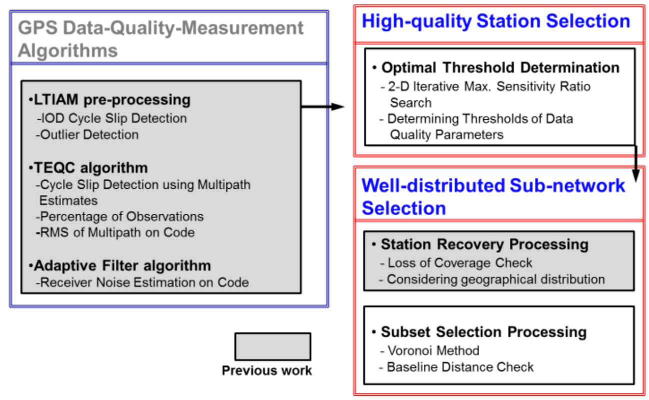

Figure 4. Optimized GNSS Network Station Selection Algorithms

3.1. DATA QUALITY MEASUREMENT ALGORITHMS

The data-quality measurement algorithms were already developed [3]. The input of these algorithms is the RINEX file collected from a station of interest, and the output is the GPS data quality information of the corresponding station. These algorithms are composed of three main parts: LTIAM pre-processing, the “Translation, Editing, and Quality Check (TEQC)” algorithm, and the adaptive filter algorithm. In the LTIAM pre-processing step, the number of IOD cycle slips, the number of outliers, and the number of short arcs are counted as separate data-quality measurements. Second, we implemented the TEQC quality metrics developed by the University Navstar Consortium (UNAVCO), which are commonly used to solve pre-processing problems [4, 5]. The TEQC method is used to obtain quality information, which includes the percentage of observations, the mean of the multipath on L1 code and L2 code, and the number of cycle slips detected using multipath estimates. Third, an adaptive filter algorithm is designed to estimate receiver noise on code measurements [6].

3.2. HIGH QUALITY STATION SELECTION

Among the GPS data quality measurements, seven quality parameters have the greatest impact on LTIAM performance. Those are the number of IOD cycle slips, number of short arcs, number of outliers, percentage of (valid) observations, multipath on L1 code and L2 code, and latency. ‘Latency’ indicates whether the number of days for which the RINEX files are properly loaded. Figures 5 and 6 show the process of determining the stations to be removed based on these seven quality parameters. Once we obtain quality measurements for all stations, we rank stations in order from worst to best for each quality parameter. The worst station is on the top and the best station is in the bottom in Figure 6. In Figure 6, the black line represents the threshold and the blue box and red box show normal stations and stations generating faulty candidates, respectively. As the threshold is reduced, both the number of stations removed, $M_{stations}$, and the number of faulty ionospheric anomaly candidates removed, $N_{candidates}$, increase. We wish to remove as many faulty candidates as possible. However, we also wish to avoid removing too many stations with acceptable data quality in order to remove only a few more faulty candidates. Thus, we measure the sensitivity ratio, λ, that expresses the relationship between the number of faulty ionospheric gradients removed and the number of stations removed:

$$\lambda = \frac{N_{candidates}(i)}{M_{stations}(i)},$$

where i is the rank of stations ($i = 1, 2, ..., \max$).

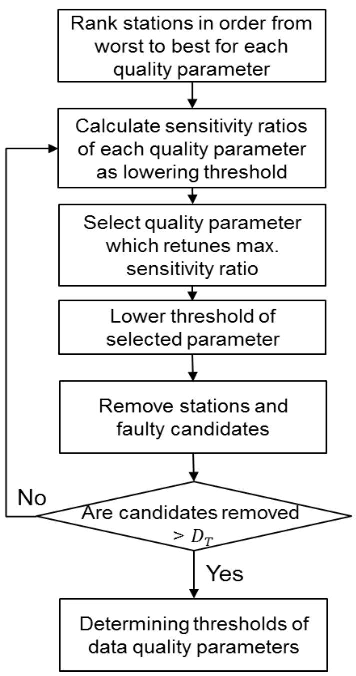

Figure 5. Flow Chart for High-Quality Station Selection Alogrithm

At this point, iteration processing starts. We compute the sensitivity ratios of each station as a lowering threshold. Next, to search for the most troublesome station and at the same time the most efficient quality parameter, we apply two dimensional concepts. In other words, the sensitivity ratios of all quality parameters are computed, and the quality parameter that returns the maximum sensitivity ratio among all quality parameters is selected. Then we lower the threshold of the selected parameter only in the current step and next remove stations and faulty candidates that exceed the lowered threshold of the selected quality parameter. This procedure is iterated until the desired numbers of faulty anomaly candidates are removed. This method explained above is the two dimensional iterative maximum sensitivity ratio search method. Because the one-step search method was developed in previous searches for troublesome stations only, whereas the proposed two-dimensional iterative search method searches for the most troublesome station and at the same time the most efficient quality parameter, the proposed method show better performance. The results of this process in CONUS are shown in Section 4.

3.3. WELL-DISTRIBUTED SUB-NETWORK SELECTION

Stations with poor quality data are determined through the two-dimensional iterative search procedure explained in the previous subsection. However, if a particular station significantly increases the geographical observability of ionospheric behavior in an area with relatively few stations, it should be retained despite poor data quality. On the other hand, the stations in dense regions do not significantly improve the geographic observability. Thus, these redundant stations, which only bring an excessive computational load, should be eliminated, even if they have high-quality data. Station recovery processing, which restores stations with poor quality data, has been developed in previous studies. In this method, first, for each station, the coverage of other stations within a 100-km radius is examined. "Station coverage" is defined as the area within which pairs of stations whose baseline is less than 100 km form. If a poor-quality station to be removed has another station nearby, the change in station coverage will be small, even after the poor-quality station is removed. However, if not many nearby stations exist, the coverage loss after discarding a station will be larger. This is the loss of coverage of a station. As a rule of thumb, if the loss of coverage after discarding a station is more than 30% of the original coverage, that station is restored despite its poor data quality.

Taking into consideration the geographical observability, eliminating redundant stations in dense regions is the final step of optimized GNSS network selection. This subset selection processing includes two methods.

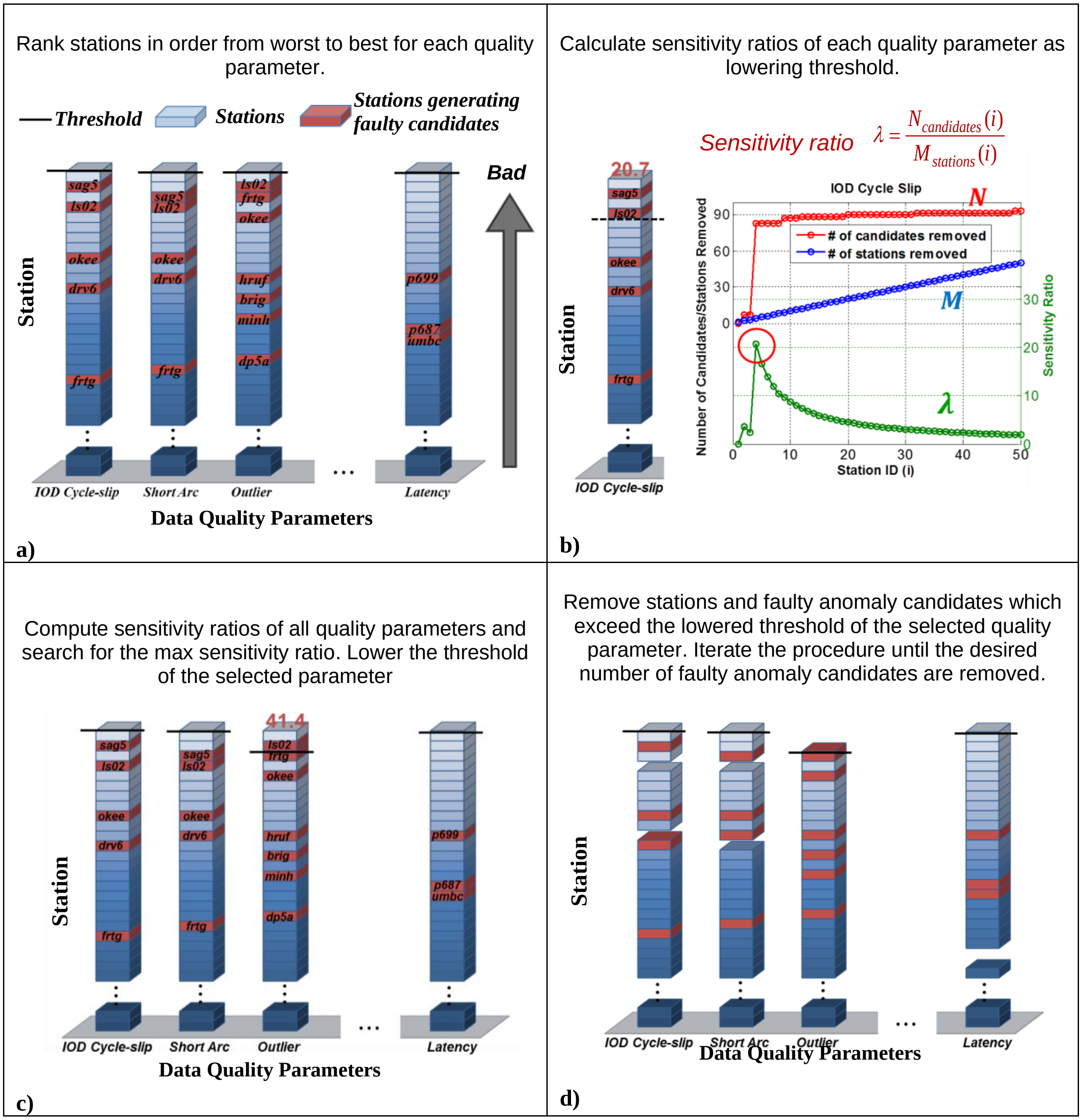

Figure 6. Steps in Determining the Threshold of Quality Parameters

One is used to make a subset of stations to be evenly distributed, and the other is used to maintain an area covered by stations with a desired baseline distance. The first method is the Voronoi method, which is a commonly-used method to divide space into a number of regions [7]. The station in a dense region has a smaller Voronoi region and, thus, to remove redundant stations, we remove the stations with the smallest Voronoi regions first, as shown in Figure 7 in the dark gray area. After a station is discarded, we redistributed the Voronoi diagram of all of remaining stations and conducted iterated processing to throw away the station with smallest Voronoi region. However, the Voroni method does not consider the loss of coverage. Thus, before discarding the station, we performed a baseline distance check. In this method, we check coverage by stations with separations of less than a desired baseline constraint. For example, if a desired baseline constraint was 40 km, as shown Figure 8, for a station, the coverage of other stations within a 40-km radius was examined. Thus, station coverage is defined as the area within which pairs of stations whose baseline is less than 40 km form. When the loss of coverage is less than 10%, the station that is chosen by the Voronoi method is discarded.

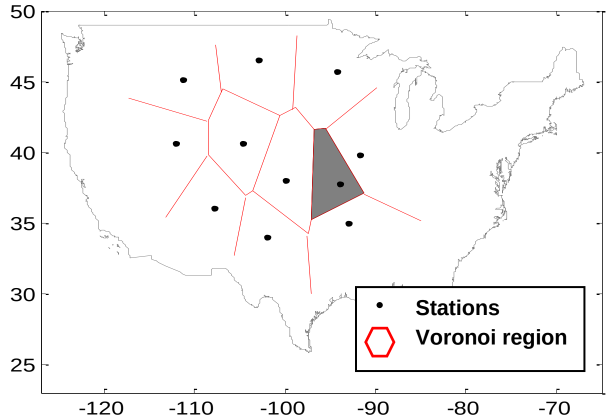

Figure 7. Overview of Stations with Voronoi Region

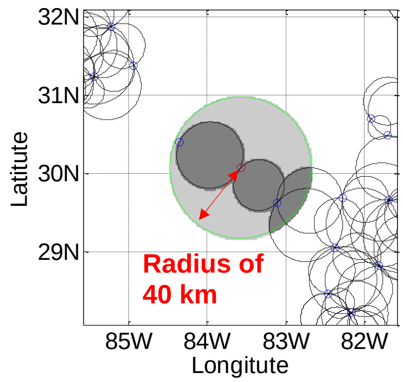

Figure 8. Check coverage by stations with separations of less than 40 km

4.0. RESULTS OF CORS STATION SELECTION

Table 1 shows the dates from which CONUS data were collected and analyzed to evaluate the performance of the optimized GNSS network selection algorithms. The geomagnetic conditions on these seven consecutive days are shown with two indices of global geomagnetic activity from space weather databases: planetary K (Kp) and disturbance storm time (Dst). In this period, a total of 1587 CORS network stations were operating in CONUS.

Table 1. Dates Analyzed to Investigate CORS Network Station Quality in CONUS

Day (UT mm/dd/yy)	Kp	Dst
24/05/12	2.0	-15
25/05/12	2.3	17
26/05/12	2.3	-6
27/05/12	1.3	14
28/05/12	2.3	23
29/05/12	2.3	23
30/05/12	2.3	16

As Kp and Dst in 오류! 참조 원본을 찾을 수 없습니다. indicate, the days on which the geomagnetic storm condition was very quiet were chosen to allow CORS station data quality to be observed while minimizing any influence of abnormal ionospheric behavior. Since it is known that an anomalous ionospheric event did not occur in this period, the threat candidates resulting from LTIAM processing are known to be faulty candidates generated by processing of poor-quality data.

The results from the GPS data-quality measurement algorithms for the station and the rank (from worst to best quality) for each quality parameter are described in [3]. The process of high-quality station selection is conducted using seven parameters of interest among the output quality parameters, and the results are shown in Figures 10 and 11.

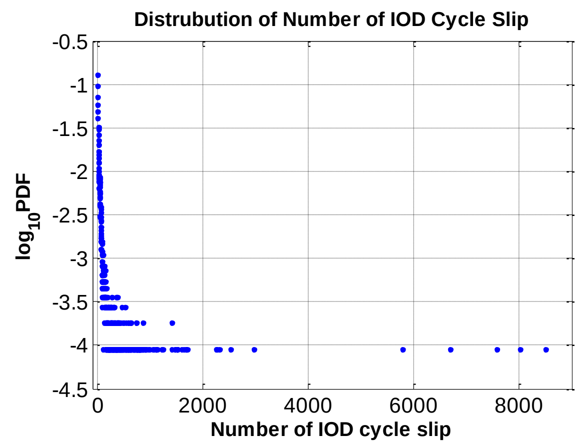

Figure 9. Probability Density Function of IOD Cycle slip for Each Station Per Day (data collected for 7 days)

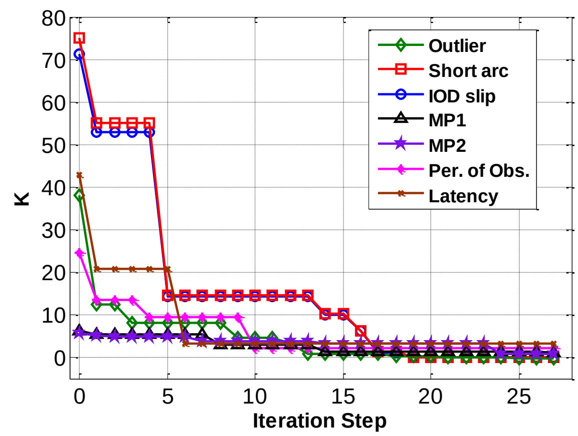

Figure 10. K Values of All Seven Quality Parameters

As shown in Figure 9, once the data from CORS stations in CONUS are collected for seven consecutive days (or longer), we then obtain the statistical distributions of each quality parameter. Data points exceeding $\mu + 9\sigma$ (the mean plus nine times the sample standard deviation) are classified as extreme outliers.

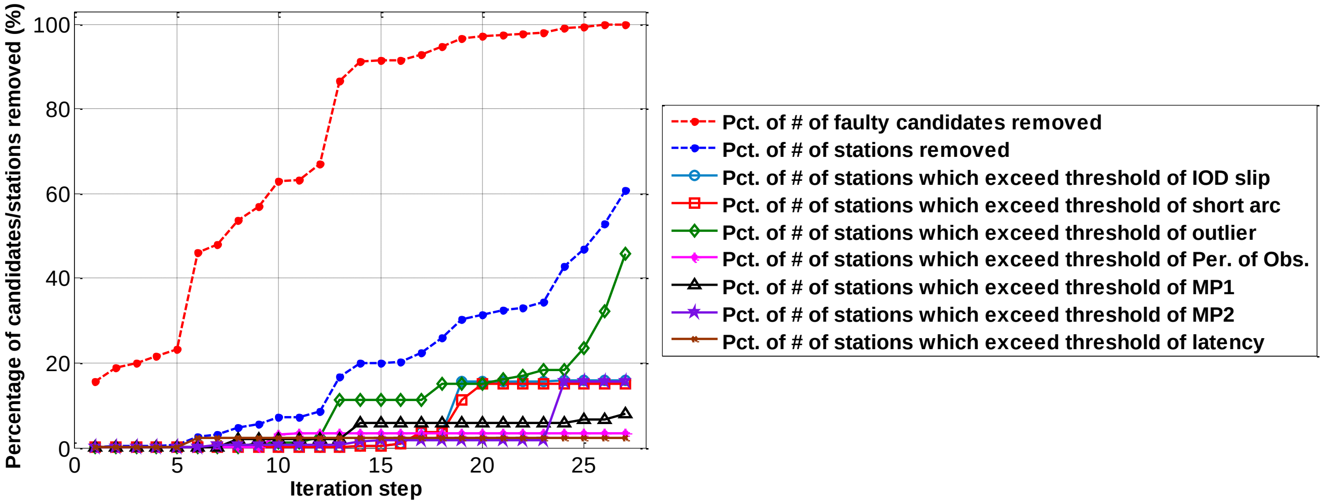

Figure 11. Percentage of Number of Stations/Candidates Removed at Each Iteration Step

After discarding these outliers, we obtain a nominal distribution for each quality parameters. Using this revised distribution, the threshold is represented as the mean plus k sigma of each parameter. Figure 10 shows the k values of seven quality parameters at each iteration step. As the steps are preceded, the threshold and k values decrease.

Accordingly, the stations are discarded with increasing frequency. Figure 11 shows the percentage of the number of faulty candidates removed as a red curve and the percentage of the number of stations removed as a blue curve at each iteration step. The other curves represent the percentage of the number of stations that exceed the threshold of the corresponding parameter. Since the number of outliers has the strongest influence on producing faulty candidates, the percentage of the number of stations that exceed the threshold of outlier, which is shown in green, is closest to the blue curve.

LTIAM processing of all stations on seven consecutive days generated 484 faulty candidates (non-existent ionospheric anomalies). As illustrated in Figure 11, 100% of faulty candidates are removed in the 27th iteration, but at a price of removing 60% of the stations. If the desired percentage of the number of faulty candidates removed was chosen as 90%, it removes only 20% of the stations, as shown in the 14th iteration. The threshold of each quality parameter at this point is shown in Table 2. Table 3 shows that this result is certainly better than that obtained previously from one-dimensional one-step search method. The two-dimensional iterative search method removes more faulty candidates, whereas fewer stations are removed. As the current method determines the threshold taking into consideration the most troublesome station and the most efficient quality parameter together unlike the previous method, the two dimensional iterative search method removes more faulty candidates, whereas the stations removed are less.

Table 2. Threshold of Data Quality Parameters for CORS Network (the desired percentage of the number of faulty candidates removed: 90%)

Quality parameter	Threshold
Number of outlier	$\mu + 0.85\sigma$
Number of IOD cycle slip	$\mu + 9.87\sigma$
Number of short arc	$\mu + 10.2\sigma$
Mean of MP1	$\mu + 1.47\sigma$
Mean of MP2	$\mu + 2.04\sigma$
Percentage of Observation	$\mu + 3.17\sigma$
Latency	$\mu + 3.35\sigma$

Table 3. Results of High Quality Station Selection on CORS Network in CONUS (based on LTIAM results from 24-30 May 2012)

	# of candidate removed on seven days (Out of 484)	# of stations removed (Out of 1587)
2D iterative search method (Current)	441 (91.1%)	316 (19.9%)
1D one-step search method (Previous)	420 (86.8%)	323 (20.4%)

We performed station recovery processing on 316 stations that were initially classified as bad stations. Among those stations, 57 stations were deemed to be "geographically critical" stations and were not removed. These 57 stations and stations removed with poor quality data are shown in green and red, respectively, in Figure 12.

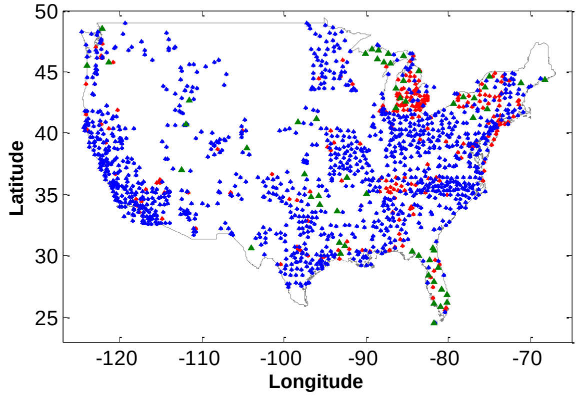

Figure 12. Map of CORS Stations in CONUS (Stations Removed in Red; Stations Restored in Green)

오류! 참조 원본을 찾을 수 없습니다.4 summarizes the results from the station recovery processing. After restoring these 57 stations, the number of faulty ionospheric anomaly candidates removed does not change much compared to the results before the station recovery processing. The high-quality station selection and the station recovery processing removed 89% of the total false anomalies while discarding only about 16% of the total stations.

Table 4. Results of Station Recovery Processing on CORS Stations Classified as "Poor Quality" (based on LTIAM results from 24-30 May 2012)

	After high-quality station selection	After station recovery
# of stations removed in CONUS (Out of 1587)	316 (19.9%)	259 (16.3%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 484)	441 (91.1%)	432 (89.3%)

Figures 13 and 14 show the Voronoi diagram of the stations that survived the high-quality station selection and the station recovery processing. The two figures show the results for subset selection when the baseline constraint is 40km and 100km, respectively. Because the number of station pairs with baselines of less than 40 km is relatively small, only about 5% of the stations in a dense area were removed by the subset selection processing, as shown in Figure 13. However, as illustrated in Figure 14 and Table 5, if baseline constraint was set to 100 km, the subset selection reduced the stations down to 46% of the original amount. This final result shows that the stations are evenly distributed and the coverage with the desired baseline distance is well maintained.

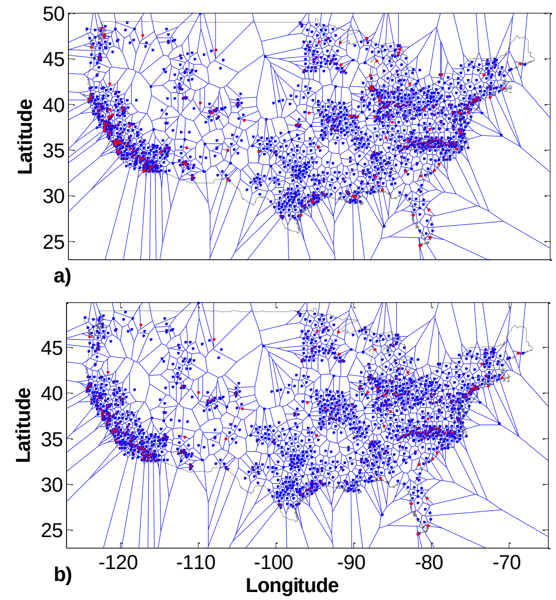

Figure 13. Map of Voronoi Diagram of CORS Stations with Baseline Constraint of 40 km: a) Before Subset Selection Processing b) After Subset Selection Processing (Baseline in Red; Voronoi Region in Blue Polygon)

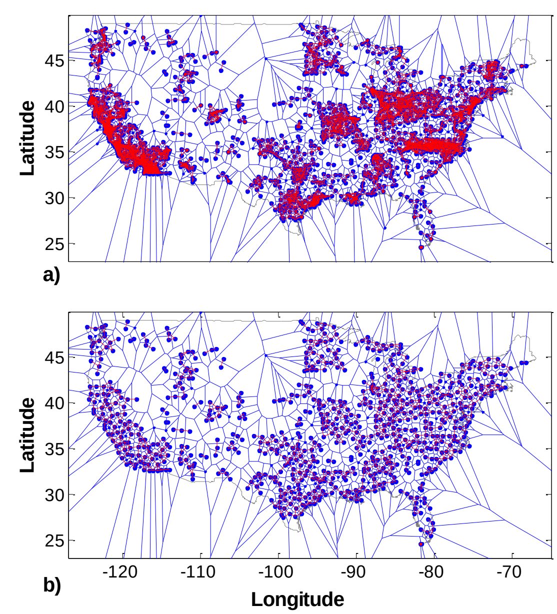

Figure 14. Map of Voronoi Diagram of CORS Stations with Baseline Constraint of 100 km: a) Before Subset Selection Processing b) After Subset Selection Processing (Baseline in Red; Voronoi Region in Blue Polygon)

Table 5. Results of Subset Selection Processing on CORS Stations Survived form Data Quality Check (based on LTIAM results from 24-30 May 2012)

	After station recovery processing	After subset selection processing
# of stations with a Baseline Constraint of 40km (Out of 1587)	1328 (83.7%)	1242 (78.3%)
# of stations with a Baseline Constraint of 100km (Out of 1587)	1328 (83.7%)	734 (46.3%)

5.0. RESULTS OF OTHER GNSS NETWORKS

To validate the performance of the high-quality, well-distributed station selection method, the proposed tool was also tested for other GNSS networks, including Japanese, European, and Canadian GNSS networks. For the Japanese GPS network, GPS earth observation network (GEONET) data was used; european reference organization for quality assured breast screening and diagnostic services (EUREF), integrated geodetic reference network of Germany (GREF), international GNSS service (IGS), and Friuli Regional Deformation Network (FReDNet) for the European GPS network; and Canadian active control system (CACS) and Canadian high arctic ionospheric network (CHAIN) data were used for the Canadian GPS network. The dates from which data were collected and analyzed to test the performance of the optimized GNSS network selection algorithms were the same dates when the CORS data was processed.

In this period, a total of 1229 GEONET network stations were operating in Japan. Among those total stations, 238 (19.4%) of all stations were chosen to be removed due to poor data quality after the high-quality station selection process. If the 238 stations selected from this quality check were removed, 479 (90.4%) of all faulty candidates out of a total of 530 faulty candidates would disappear, as shown in **오류! 참조 원본을 찾을 수 없습니다.**6. These results are similar with the case of the CORS network. This indicates that the qualities of the two networks are also similar.

Table 6. Results of Station Recovery Processing on GEONET Stations Classified as "Poor Quality" (based on LTIAM results from 24-30 May 2012)

	After high-quality station selection	After station recovery
# of stations removed in GEONET (Out of 1229)	238 (19.4%)	236 (19.2%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 530)	479 (90.4%)	477 (90.0%)

Figure 15 shows result of the station recovery process for GOENET. Since GEONET forms a dense network in comparison to the CORS network, only two stations were considered to be 'geographically critical' stations among the 238 stations with poor-quality data. Also in the case of GEONET, since there are many pairs of stations with baselines less than 40 km, as shown in Figure 16 and Table 7, the number of stations could be reduced to approximately 40% of the total through subset selection process with a baseline constraint of 40 km. Therefore, we need to process only about 40% of GEONET stations and yet maintain the similar level of ionospheric obervability.

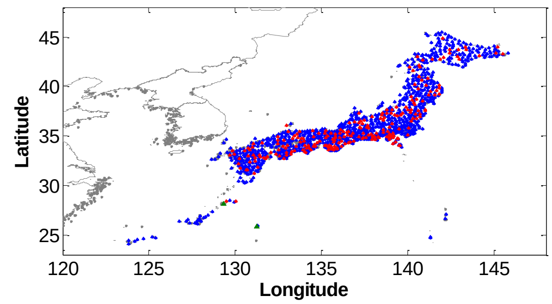

Figure 15. Map of GEONET Stations in Japan (Stations Removed in Red; Stations Restored in Green)

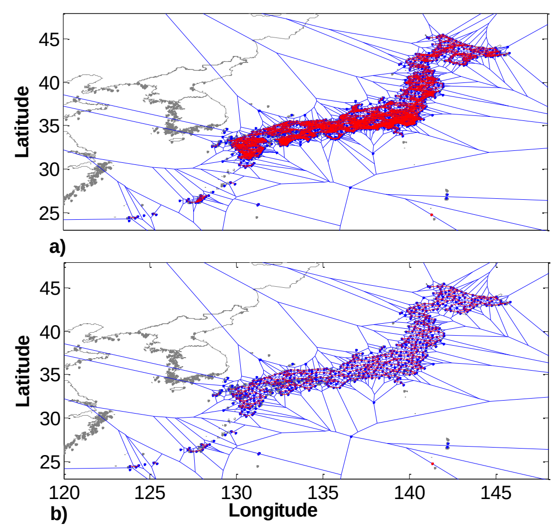

Figure 16. Map of Voronoi Diagram of GEONET Stations with Baseline Constraint of 40 km: a) Before Subset Selection Processing b) After Subset Selection Processing (Baseline in Red; Voronoi Region in Blue Polygon)

Table 7. Results of Subset Selection Processing on GEONET Stations Survived form Data Quality Check (based on LTIAM results from 24-30 May 2012)

	After station recovery processing	After subset selection processing
# of stations with a	993 (80.8%)	473 (38.5%)

Baseline Constraint of 40km (Out of 1229)		

The results from European and Canadian GPS networks are shown respectively in Figures 17 and 18. In the case of these two networks, the test was performed only up to station recovery processing as there are not too many stations. As shown in Table 8, for the European GPS networks, 29(12%) stations out of the total 246 stations are removed and through this approximately 94% of faulty candidates are eliminated. Furthermore, for the Canadian GPS network, only 7(16%) stations of the total 45 stations are removed so 100% of all faulty ionospheric anomaly candidates are removed as shown in Table 9.

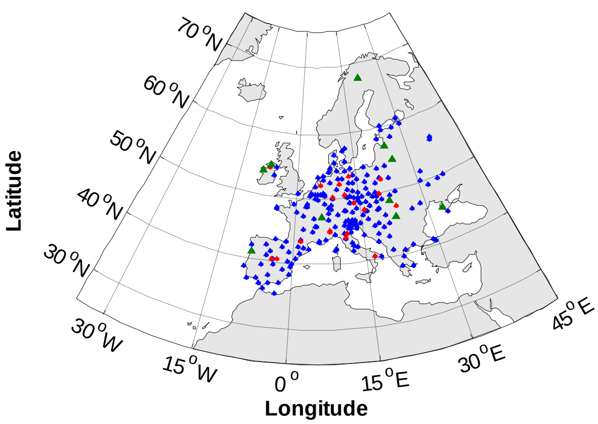

Figure 17. Map of European Stations (Stations Removed in Red; Stations Restored in Green)

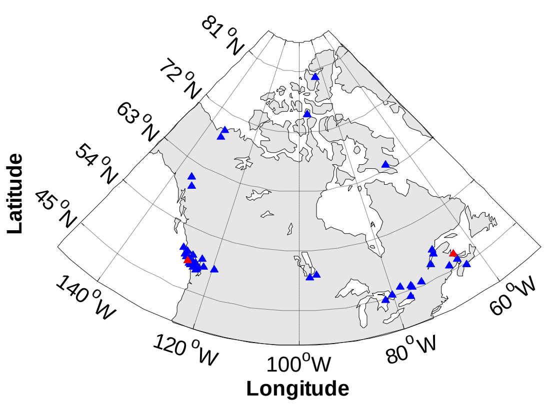

Figure 18. Map of Canadian Stations (Stations Removed in Red; Stations Restored in Green)

Table 8. Results of Station Recovery Processing on European Stations (based on LTIAM results from 24-30 May 2012)

	After station recovery
# of stations removed in European (Out of 246)	29 (11.8%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 32)	30 (93.8%)

Table 9. Results of Station Recovery Processing on Canadian Stations (based on LTIAM results from 24-30 May 2012)

	After station recovery
# of stations removed in Canadian (Out of 45)	7 (15.6%)
# of faulty ionospheric anomaly candidates removed on seven days (Out of 48)	48 (100%)

Since the results of other GNSS network show similar or improved performance compared to the results of the CORS network, it was concluded that the optimized GNSS network selection proposed in this study exhibits enhanced performance.

6.0. SUMMARY

This paper presents the results of the optimized method to select GNSS network stations with well-distributed, high-quality data and verify the performance of proposed method. Specific stations with poor quality data produce many undesired faulty candidates in LTIAM, and cause excessive processing time in the manual validation. The optimized method developed in this paper achieved a 89% reduction in faulty anomaly candidates while discarding only 16% of CORS stations. The results from processing recently-collected GNSS data confirm the proposed methods also perform well for other GNSS networks. Geographical observability of the ionosphere with a baseline constraint of 40 km can be maintained by processing only 40% of the GEONET stations. Consequently, this method greatly reduces excessive processing time and effort in the manual validation process by faulty candidates and reduces unnecessary processing time resulting from processing redundant station data, where this should benefit the future to building ionosphere threat models for all regions so that GBAS will be fielded in the future.

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